

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int at[10], bt[10], pr[10];
6      int n, i, j, temp;
7      int time = 0, over = 0;
8      int sum_wait = 0, sum_turnaround = 0;
9      int start, end;
10
11     float avgwait, avgturn;
12
13     printf("Enter the number of processes\n");
14     scanf("%d", &n);
15
16     for(i = 0; i < n; i++)
17     {
18         printf("Enter the arrival time and execution time for process %d\n", i + 1);
19         scanf("%d %d", &at[i], &bt[i]);
20         pr[i] = i + 1;
21     }
22
23     // Sort by Arrival Time
24     for(i = 0; i < n - 1; i++)
25     {
26         for(j = i + 1; j < n; j++)
27         {
28             if(at[i] > at[j])
29             {
30                 temp = at[i]; at[i] = at[j]; at[j] = temp;
31                 temp = bt[i]; bt[i] = bt[j]; bt[j] = temp;
32                 temp = pr[i]; pr[i] = pr[j]; pr[j] = temp;
33             }
34         }
35     }
36
37     printf("\nProcess |Arrival time |Execution time |Start time |End time |Waiting\n");
38
39     while(over < n)
40     {

```

```

41 // CPU idle handling
42 if(time < at[over])
43     time = at[over];
44
45 start = time;
46 end = start + bt[over];
47
48 printf("p[%d]\t\t |%d\t\t\t |%d\t\t\t |%d\t\t\t |%d\t\t |%d\n",
49 pr[over],
50 at[over],
51 bt[over],
52 start,
53 end,
54 start - at[over]);
55
56 sum_wait += (start - at[over]);
57 sum_turnaround += (end - at[over]);
58
59 time = end;
60 over++;
61 }
62
63 avgwait = (float)sum_wait / n;
64 avgturn = (float)sum_turnaround / n;
65
66 printf("Average waiting time is %f\n", avgwait);
67 printf("Average turnaround time is %f\n", avgturn);
68
69 return 0;
70 }

```

Enter the number of processes

3

Enter the arrival time and execution time for process 1

1 3

Enter the arrival time and execution time for process 2

2 6

Enter the arrival time and execution time for process 3

3 8

Process	Arrival time	Execution time	Start time	End time	waiting
p[1]	1	3	0	3	-1
p[2]	2	6	3	9	1
p[3]	3	8	9	17	6

Average waiting time is 2.000000

Average turnaround time is 7.666667

Process exited after 20.18 seconds with return value 0

Press any key to continue . . . |